

[Home](#) > [My Courses](#) > [Financial Data](#) >[M5: News Data and Sentiment Analysis](#) > Lesson Notes

Lesson Notes

MODULE 5 | LESSON 4

CRITICAL THINKING TO ASSESS DATA VERACITY

Reading Time	15 minutes
Prior Knowledge	All Financial Data content, Financial Markets Module 7 content
Keywords	Information Literacy, Fake News

In this lesson, we examine the characteristics of critical thinking. We'll look at a systematic literature review that addresses how critical thinking skills were used to determine if news was "fake news."

1. Critical Thinking

Critical thinking is not just something we'll discuss once. As Ismail writes, "Critical thinking involves the purposeful, goal-directed thinking during the process of making decisions based on the evidence in the scientific problem-solving process" (1). Throughout this program, you'll be expected to utilize critical thinking skills for a variety of purposes, but here, we'll focus on using critical thinking to determine whether a source is reliable.

The article by Ismail addresses the importance of critical thinking skills for getting hired and being productive once employed. The article highlights a study that was done on employees who had five years of experience in an engineering job, particularly mechanical engineering. The period of five years is considered to be a starting point for building maturity in one's career.

In accordance with this study, the following questionnaire was given to mechanical engineers with 5 years of work experience. The questions relate to:

- Problem identification in a complex system
- Improving explanations, analysis, and evaluation skills
- Searching for alternative solutions
- Searching for innovative solutions
- Justifying decisions based on strong evidence
- Ability to persevere to accomplish responsibilities assigned
- Ability to understand and adapt to the culture of the community

Critical thinking skills are highly desirable in the workforce because they're necessary for working effectively on a team, analyzing data, crafting arguments, making decisions, and solving problems. As such, it's important to strengthen this skill as much as you can throughout the program.

Your critical thinking skills will be evaluated in a variety of ways—from forum posts to group work projects to your capstone project. One critical thinking skill we focus on here is information literacy, which is essential as you search for valid and credible sources to find the data you need, to support and develop your ideas, and more. These skills are incredibly important in financial engineering jobs as well as other STEM-related professions. Later in this lesson, we'll apply critical thinking skills to determine whether news is fake or credible.

2. Information Literacy

The American Library Association defines information literacy as “a set of abilities requiring individuals to recognize when information is needed and have the ability to locate, evaluate, and use effectively the needed information.” As such, information literacy means that you first know how to research and find the sources you're looking for and how to evaluate whether a source is a credible and reliable source of information. While this may sound simple, it has become an increasingly difficult task with the advent of fake news, crowdsourced information, and anonymous internet authors.

As such, the first thing we'll have you do is complete two modules from the Information Literacy tutorial, as outlined in

the Required Readings for this lesson. Please be sure to complete the following modules:

- TU2: Identify the Required Resources
- TU5: Evaluate the Resources

Given the vast amount and variety of information on the internet, evaluating online resources is more important than ever. Can you trust information from a blog? What about using information that was posted as an answer to someone's question?

As you prepare yourself to get data and research for your empirical studies, you may encounter news articles and headlines that you use in a sentiment analysis or text analysis. Be warned that there is plenty of fake news on the internet.

3. Unreliable Sources

Journalism has changed drastically over the last few decades with the rise of the internet, smartphones, and social media. The development of 24-hour news networks has encouraged cable news to embrace sensationalist headlines while promoting the partisan viewpoints of its owners and balancing the needs and demands of advertisers. TV and social media have overtaken print as sources of news, so newspapers and local journalism have struggled to survive. As a result, people must look to other sources for their news, and oftentimes, these sources may have conflicts of interest that must be considered when evaluating whether their reporting is reliable and credible. There's a lot of pressure in journalism to break the story first, regardless of whether reporters have all the facts. As a result, initial reports are often wrong, and networks or publications may find themselves issuing corrections or retractions later.

These are all important realities to keep in mind when finding sources on the internet. This isn't to say that all cable news is unreliable or that you should never trust a blog. Instead, the takeaway should be to do your research. When deciding whether or not a source is credible, ask a few key questions:

- Who wrote this text? Do they have a reputation as a credible scholar or expert in this field or on this topic? Is this a reliable organization that has demonstrated a commitment to getting

the facts right? Is there any potential political bias or conflict of interest you should be aware of?

-When was this text written? Is the source outdated or irrelevant, considering the topic you're researching? How does the source respond to or incorporate modern developments pertaining to the topic?

-Does the text's argument, logic, and reasoning hold up? Can the information in the text be easily disproven? Can most facts or statistics be substantiated by another source? Does the author cite reliable sources as evidence?

-What is the style or tone of the text? Is it more academic and objective, or is it emotional and provocative? Is the text trying to persuade you to buy/use/read/watch/do something?

As you complete the information literacy tutorial, you'll learn the skills and strategies you need to evaluate a source and answer these questions.

Suppose you were building a trading strategy using machine learning algorithms that perform sentiment analysis. These methods require headlines, news articles, and perhaps even social media posts. The existence of fake news could turn these models into garbage in, garbage out boxes. Alternatively, perhaps in doing your capstone project, you decide to use a source that turns out to be fake. This could have disastrous implications for your research findings. In our age of misinformation, it is more important than ever to be able to identify (and avoid) unreliable sources.

4. Critical Thinking about Fake News

Please read Machete and Turpin's "The Use of Critical Thinking to Identify Fake News: A Systematic Literature Review." The article reviews the literature on how critical thinking can be used to identify fake news. The articles that explicitly discussed critical thinking found that most students do not have sufficient critical thinking skills to identify news that is fake.

Once you have read the article, apply the information you have learned from the information literacy course to determine whether something is fake news.

1. Is the information accurate, as determined by checking other sources?
2. Is the article written by an authority? Is the authority recognized by peers?
3. Is the information complete? Take care to note stories that omit key information.
4. Is the story timely? Have there been updates or other news that indicate the information may be outdated?
5. Was the information presented as clickbait, merely to attract an audience, or is the information provided as a genuine news story?

Suppose in your due diligence, you answer some of the above questions in the negative. The question to ask then is "has this story been repeated elsewhere, and has it been confirmed or verified by reliable sources?" Any doubts on your part could then prevent that article from becoming part of the data for your sentiment analysis.

5. Conclusion

Clearly, creating fake news would be a violation of good ethical practice. As an FE, you can become more information-literate by learning how to assess the quality, authority, comprehensiveness, and timeliness of information you find on the web. This information may become data that you use to build and train your models for FE. Building information literacy will actually help promote good ethical practice as well.

In concluding this module, let's remember to be honest and vigilant in avoiding the creation, spread, or inclusion of fake news in your studies. Even if you use AI to determine what is fake, be sure that you confirm what gets used. The saying "garbage in, garbage out" applies, but the situation can be worse if we assume fake stories to be true and make key decisions based on false information.

References

- Alma Mater Studiorum. *Information Literacy Tutorial*. Università di Bologna, 2018.
<https://sba.unibo.it/en/services/training/tutorial-di-information-literacy-1>.

- Hattingh, Marié, et al. "The Use of Critical Thinking to Identify Fake News: A Systematic Literature Review." *Responsible Design, Implementation and Use of Information and Communication Technology*, 2020, pp. 235–246, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7134234/>.
- Ismail, W. Omar Ali Saifuddin Wan, et al. "The Essential of Engineering Education Involving Critical Thinking and Problems Solving Skills among Mechanical Engineer Employees." *IOP Conference Series: Materials Science Engineering*, vol. 697, 2019, <https://iopscience.iop.org/article/10.1088/1757-899X/697/1/012017>.

Copyright 2024 WorldQuant University. This content is licensed solely for personal use. Redistribution or publication of this material is strictly prohibited.

